

Rayleigh Fingerprint-Based DAS over a 688-km Repeatered Link with Co-Propagating 15-Tb/s WDM Transmission

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Abstract We report the first demonstration of distributed acoustic sensing based on cross-correlation of the fiber's Rayleigh fingerprint over >600 km using 500-MHz bandwidth pulses. The link includes 90 km of field fiber for seismic monitoring. ©2026 The Authors

Introduction

Rayleigh-based distributed acoustic sensing (DAS) can transform the existing fiber infrastructure into a dense array of strain sensors for large-scale seismic monitoring. To enable this, recent work has focused on improving sensitivity [1, 2], extending operation to repeatered links [3–10], and achieving coexistence with telecom channels [3, 7, 8].

To date, long-range DAS over repeatered links has mainly relied on pulse-compression (PC) schemes with phase demodulation [11], using tens-of- μ s-long pulses to increase pulse energy combined with frequency diversity over <250 MHz bandwidth [3–10]. Perturbations are retrieved by computing phase differences between z-locations separated by the gauge length. Consequently, these schemes suffer from fading points and phase unwrapping errors, which may degrade the sensor sensitivity. To address these limitations, [1] proposed a DAS scheme based on wider-bandwidth chirps, ranging from 400 MHz up to 5 GHz in more recent works [12]. The resulting backscattered trace, rich in spectral content, is commonly referred to as a Rayleigh fingerprint [13, 14]. In these settings, the sensing task is transformed into a time-delay estimation problem: strain

induces a local shift of the fingerprint, which can be tracked robustly via cross-correlation. Although fingerprint-based DAS has been demonstrated over single spans up to 80 km [15], its extension to repeatered links remains unexplored. In the scheme of [1], the spatial resolution is set by pulse duration; therefore, achieving meter-scale resolution requires sub- μ s pulses. This limits the pulse energy and hinders telecom co-propagation due to cross-phase modulation (XPM) [3]. A technique to employ longer, higher-energy pulses was introduced in [16], but validated only over 10 km.

Here, we report the first results of Rayleigh fingerprint-based DAS over a >600 km repeatered link with 14 spans. This is achieved using a partial PC scheme with a novel phase-noise compensation method, which enables cross-correlation with chirped pulses of 30 μ s duration and 500 MHz bandwidth. The system is validated through controlled lab perturbations and measurement of a M_w 4.5 earthquake in the field-deployed section of the link, showing improved sensitivity compared to phase-based methods. We also demonstrate compatibility of the DAS operation with co-propagating telecom channels in the same fiber, achieving 15.1 Tb/s data rate.

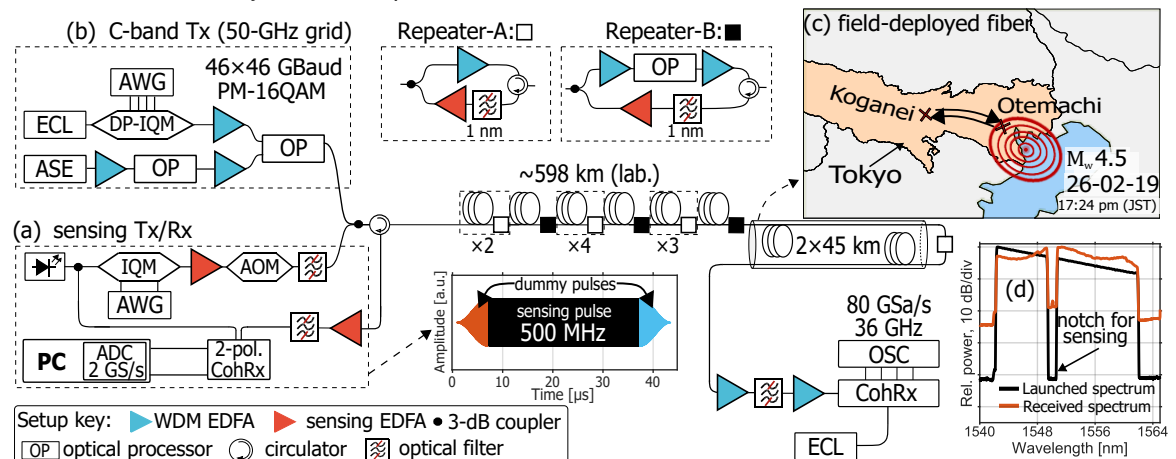


Fig. 1: Setup for DAS with WDM transmission. (a-b) Schematic of Tx/Rx blocks, repeater units, and fiber link; (c) Map showing field-deployed link route with earthquake epicenter location. (d) Tx/Rx spectra measured at 0.1 nm resolution bandwidth.

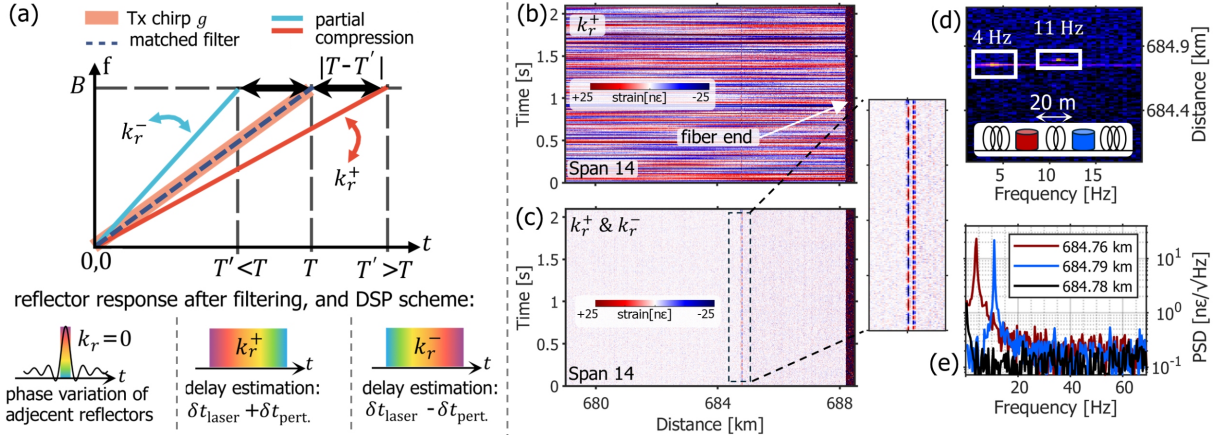


Fig. 2: DAS validation in lab link. (a) Schematic of the proposed residual-chirp pulse-compression scheme. (b) Strain map from time-delay estimation on the fingerprint obtained from k_r^+ . (c) Strain map after compensation with the delays obtained from k_r^- : $[\delta t(k_r^+) - \delta t(k_r^-)]/2$; apparent strain induced by intra-pulse laser frequency noise is mitigated, while the strain signal is retained. (d-e) Validation of sensor spatial resolution.

System Description

The experimental setup for DAS with WDM channels co-propagation is shown in Fig. 1. The sensing signal at 1550.12 nm [Fig. 1 (a)] was generated from a <100-Hz-linewidth laser using a single-polarization I/Q modulator (IQM), producing chirped pulses with 30 μ s duration, 500 MHz bandwidth and a 6.9 ms repetition period. Smooth dummy pre- and post-pulses with 7.5 μ s rise/fall times were added at a different frequency compared to the chirped pulse to reduce XPM effects on telecom channels [3]. The backscattered signal was detected by a dual-polarization heterodyne coherent receiver (CohRx) and digitized at 2 GS/s (12-bit), with continuous data streaming to computer storage. The digital signal processing (DSP) is explained in the next section.

The telecom transmitter [Fig. 1(b)] emulated 50-GHz-spaced WDM channels using amplified spontaneous emission (ASE) noise spanning 1542–1562 nm, and generated a 46 GBaud PM-16QAM tunable test channel. Two cascaded optical processors (OPs) shaped the ASE spectrum and carved a fixed 1.2-nm-wide notch for the DAS channel and a tunable 0.4-nm-wide notch for the test channel. Cascading the OPs reduced the notch roll-off, allowing to emulate neighboring-channel interference with ASE, providing a conservative estimate of telecom performance [17]. The received signal was detected by an intradyne CohRx and digitized at 80-GS/s by a real-time oscilloscope (OSC) triggered by the sensing AWG to assess XPM effects. The receiver DSP followed [18].

Sensing and telecom channels were combined with a 3-dB coupler and launched into the 688 km link at 15 dBm total average power (telecom) and 2 dBm peak power (sensing). The link comprised 14 spans: 12 lab spans of 40–51 km and 2 $\times$ 45 km field spans [see Fig. 1(c)]. Two types of EDFA-based repeaters were used: Repeater-A and

Repeater-B. In both, bidirectional operation was achieved using a 3-dB coupler at the repeater input and a circulator at the output. Repeater-B also included an OP in the forward path to counteract the spectral tilt accumulated along the link. An 8-dB pre-tilt was applied to the launch spectrum as shown in Fig. 1(d).

Correlation-based DAS

In correlation-DAS, a chirped pulse g , with bandwidth B , duration T , and chirp rate $k = B/T$, is launched into the fiber [16]. After coherent detection, the backscattered field is digitally filtered by h . Full PC is achieved when $h = g$ [11], yielding a spatial resolution $S_r = v_g/2B$ (v_g is the group velocity). To enable correlation-DAS [16], partial PC is implemented using h with bandwidth B but duration $T + \Delta T$. This yields the fingerprint with resolution $S_r = v_g/2B + 3/8v_g|\Delta T|$, equivalent to using a short chirped pulse of duration $\propto |\Delta T|$ and bandwidth B . Nevertheless, in long pulses (e.g., $T=10$ – 30μ s, used to increase pulse energy) intra-pulse laser phase noise is converted by the residual chirp into time jitter of the fingerprint.

To address this issue, we propose the novel method sketched in Fig 2(a). For a given target resolution S_r , the sign of ΔT selects two symmetric partial PC configurations that have residual chirp rates $k_r^\pm \approx \pm k|\Delta T|/T$ of opposite sign. The transmitted chirp is identical in both configurations; only the digital chirp rate changes sign. As a result, the laser noise contribution remains common-mode, whereas the strain-induced delay reverses sign. Subtracting the delay maps mitigates the laser term while retaining the strain signal. To validate the method, we replaced the field fiber with lab fiber including two piezoelectric transducers (PZTs) separated by 20 m. The strain map obtained for k_r^+ [Fig. 2(b)] shows slow spatial fluctuations due to intra-pulse laser phase noise. After compensation with k_r^- , the noise is suppressed and the perturbations are clearly visible.

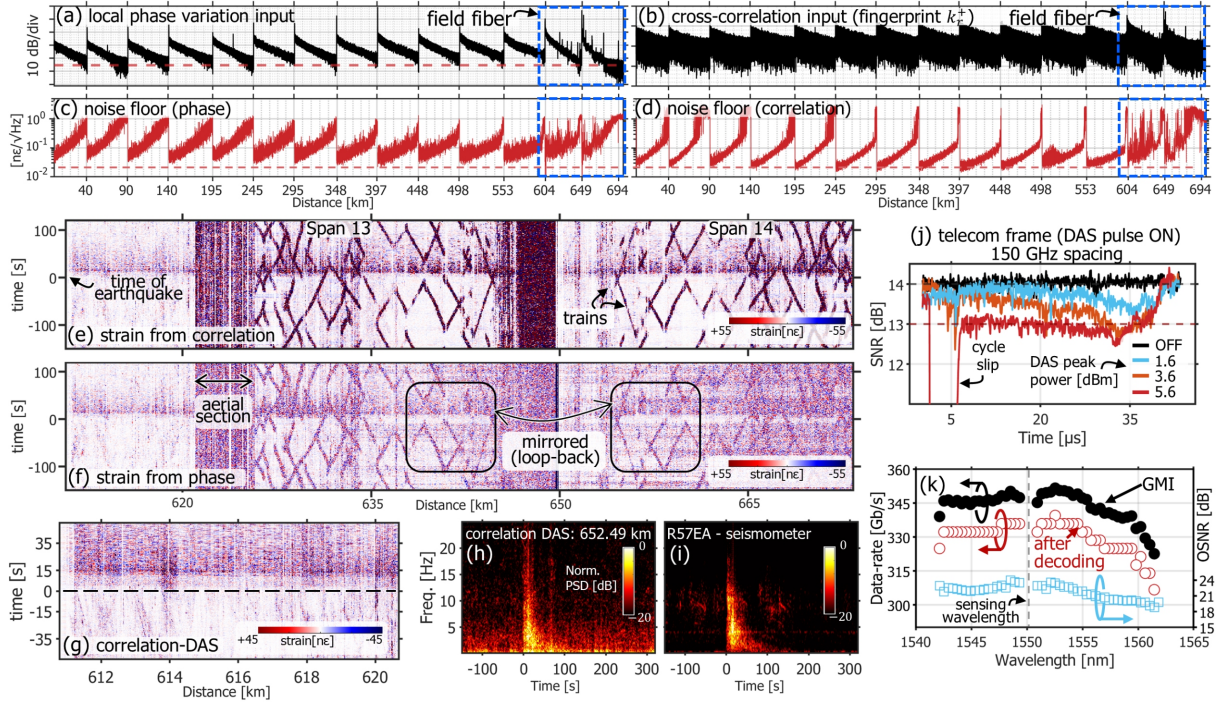


Fig. 3: (a) Traces after longitudinal coherent averaging (phase-DAS). (b) Fingerprint k_r^+ for correlation-DAS. (c-d) Sensor noise floor. (e-f) Acoustic events in the field fiber (diagonal traces are from passing trains). (g) Close-up at the time of the M_w 4.5 earthquake. (h-i) DAS versus seismometer spectra. (j) Telecom SNR versus DAS peak power. (k) Data-rate and OSNR of the 46 WDM channels.

ble in Figs. 2(c–e), with no detectable spatial cross-talk. Time-delay estimation was implemented as in [19].

Results and Discussion

We compared the proposed correlation-DAS with conventional PC phase-DAS. Figures 3(a–b) show the amplitude of the backscattered traces used for local phase-variation and cross-correlation processing, respectively. For phase-DAS [Fig. 3(a)], we applied longitudinal coherent averaging [7], exploiting the pulse frequency content to mitigate fading points at the cost of worsened spatial resolution. We used $N_z = 315$ spatial samples within the gauge length. With a spatial sampling step $\delta z \approx 5$ cm (from 2 GS/s sampling rate) and a resolution after matched filtering of $\Delta z \approx 20$ cm (from 500 MHz bandwidth), the effective spatial resolution is $S_r = N_z \delta z + \Delta z \approx 16$ m. For correlation-DAS, we set $S_r = 16$ m, which gives $\Delta T \approx 200$ ns ($T = 30$ μ s, $B = 500$ MHz). Figure 3(b) shows the trace obtained with chirp rate k_r^+ ; here, fading points act as fingerprint features for cross-correlation, so no additional DSP for mitigation was applied. The noise floor along the link is shown in Fig. 3(c–d) for both DAS schemes; correlation-DAS achieves a generally lower noise floor since it avoids direct phase extraction, making the DAS sensitivity independent of the fading-mitigation algorithm. Figures 3(e–g) show the strain maps for the field spans during a M_w 4.5 earthquake [epicenter relative to the fiber shown in Fig. 1(c)]. At the end of Spans 13 and 14, low

SNR causes phase-unwrapping errors in phase-DAS, which are not observed in correlation-DAS. The measured frequency content closely matches that of a nearby citizen seismometer (Raspberry Shake R57EA) [Fig. 3(h–i)]. For telecom measurements [Fig. 3(j–k)], the oscilloscope trigger was provided by the sensing pulse AWG and delayed by the link time-of-flight to capture the test channel affected by pulse co-propagation. Cycle slips in carrier-phase recovery were observed for DAS peak power > 5 dBm [Fig. 3(j)]. We therefore set the DAS peak power to ≈ 2 dBm, limiting the XPM-induced SNR penalty to < 1 dB. We estimated the data rate of the 46×46 -GBaud channels using the generalized mutual information (GMI) and implemented a decoding scheme [18] using low-density parity-check codes. We achieved 15.1 Tb/s, with per-channel data rate > 300 Gb/s [Fig. 3(k)].

Conclusions

We reported the first demonstration of correlation-based DAS over a > 600 km repeatered link using 500-MHz chirped pulses. The proposed DAS implements partial pulse-compression with a novel phase-noise compensation scheme to extract strain information from time-delay estimation on the Rayleigh fingerprint. The scheme shows improved sensitivity compared to phase-DAS. We also demonstrate its coexistence with 46 WDM channels, achieving 15.1 Tb/s data-rate. These results pave the way for high-sensitivity, long-range DAS systems compatible with the existing telecom fiber infrastructure.

Acknowledgements

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